

AI WORKOUT TRACKER

Project Assessment & Strategic Roadmap

A full-stack AI fitness companion combining workout logging, live phone health data, and a multi-model conversational AI coach that understands your real training, body metrics, and medical conditions.

Frontend: React Native · Expo · TypeScript · Zustand · NativeWind | **Backend:** Node · Express · Prisma · MongoDB · JWT
| **AI:** Claude · GPT · Gemini · DeepSeek

Prepared for mentor review

1. Performance Matrix — Feature by Feature

Each feature scored 1–5 across four axes. **C**=Completeness, **R**=Reliability, **UX**=Polish, **E**=Competitive Edge. Ratings reflect the real state of the code — kept honest on purpose.

Feature	C	R	UX	E	Status & Honest Note
Authentication					JWT + bcrypt, forgot-password. Solid & standard. Phone-auth scaffolded, not finished.
Workout Tracking					Live sets/ reps/ weight, timer, history, delete. The core — works well.
Exercise Library					20 exercises with real photos, step tutorials & YouTube. Newly upgraded.
Programs					Preset PPL / 5×5 + custom program builder + start-from-program flow.
Health Dashboard					Real, body-personalized steps/calories/distance + activity rings. Device-only; estimates.
AI Coach					Multi-model, data-aware, medical-aware. Biggest edge — but reliability is the #1 fix.
Medical Awareness					Injuries feed AI safety advice. Genuinely uncommon; underused today.
Notifications					Scheduled motivational pushes. Good for retention.
Design System					Dark/light themes, consistent NativeWind UI. Presents very well.
Backend					Express + Prisma + MongoDB. Gaps: secrets-in-repo, no tests.

Takeaway: A strong, polished full-stack app with a genuinely differentiated AI + health layer. The technical foundation is done — the next phase is hardening reliability and deepening the AI's intelligence.

2. How We're Different — Competitive Positioning

Competitor	Their Strength	What They Lack That We Have
Hevy / Strong	Best-in-class workout logging	No AI coaching · No phone health integration
MyFitnessPal	Huge nutrition & calorie database	Weak strength logging · No conversational AI
Fitbod	AI-generated workout plans	No chat coach · Not aware of your injuries
Apple Fitness+	Activity rings & health data	No structured strength logging · No AI coach

■ Our Wedge — the “all-in-one intelligent coach”

We're the only one combining **workout logging + live phone health data + a multi-model conversational AI** that knows your real lifts, body metrics, daily activity, **and** medical conditions. The **injury-aware safety angle** (“given your knee injury, I avoided heavy-knee-load exercises”) is something none of the big players do — that's the headline.

3. Future Improvements — Prioritized Roadmap

P0

Reliability — do first (demo-blocking)

- Add valid AI API keys + move secrets out of the repo into secure env; show an explicit “offline mode” badge instead of silent fallback.
- Add age + activity level to the profile → accurate TDEE (age is currently assumed).
- Integrate Apple HealthKit / Google Fit for true steps, heart rate & sleep (vs pedometer-only estimates).

P1

Depth & Stickiness

- Progress analytics: volume trends, automatic PR detection, 1RM estimates, strength-over-time charts.
- Streaks & gamification, rest timer, plate calculator.
- AI memory (RAG over workout history) so the coach references past weeks.

P2

Expansion

- Nutrition logging + barcode / photo calorie scan.
- Wearable companion (Apple Watch / Wear OS).
- Social: share workouts, leaderboards, coach-assigned programs.
- Progress photos + body-measurement tracking.

ENG

Engineering Hygiene (shows maturity)

- Automated tests + CI, error monitoring (Sentry).
- API rate-limiting, response caching for AI, product analytics.

Closing line for the meeting: “The foundation is built and polished. The differentiated bet — an injury-aware, data-driven AI coach — is what we double down on next.”